Do Now · retrieval

Introduction to time and space complexity

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What makes a binary search trace convincing?

Q2

State the average-case time complexity of quick sort for n items.

Q3

How many values are in an int array of length n?

Q4

Apply an eight-bit left shift by two to 00010110_2 and bitwise AND to 00110110_2 with 00001111_2. Give both results in binary.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Analyse the loops — One full loop runs n iterations; a pair of n-sized nested loops performs n^2 inner iterations.

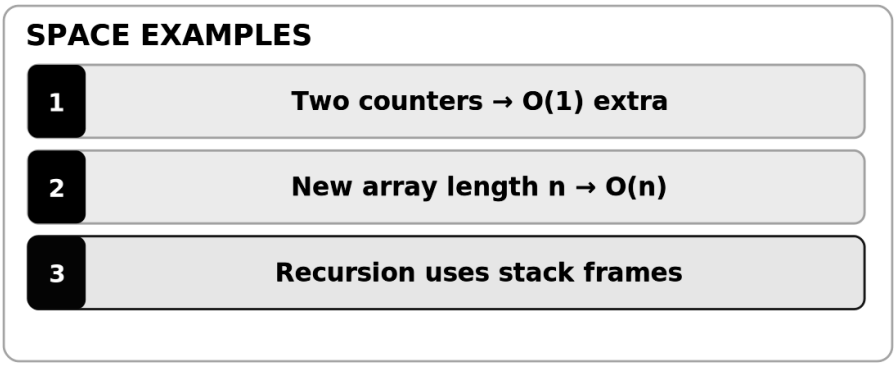
Core idea 2

Separate time and space — Big O time counts growth of work; auxiliary space measures extra storage beyond the original input.

Core idea 3

Compare feasible scales — At n=16, exponential growth can be dramatically more expensive than linear growth. Big O omits constants and measured run time.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

time complexity — Asymptotic growth in algorithm work.

auxiliary space — Extra memory used by an algorithm.

Big O — Upper asymptotic bound commonly used to classify growth.

quadratic — Growth proportional to n^2.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

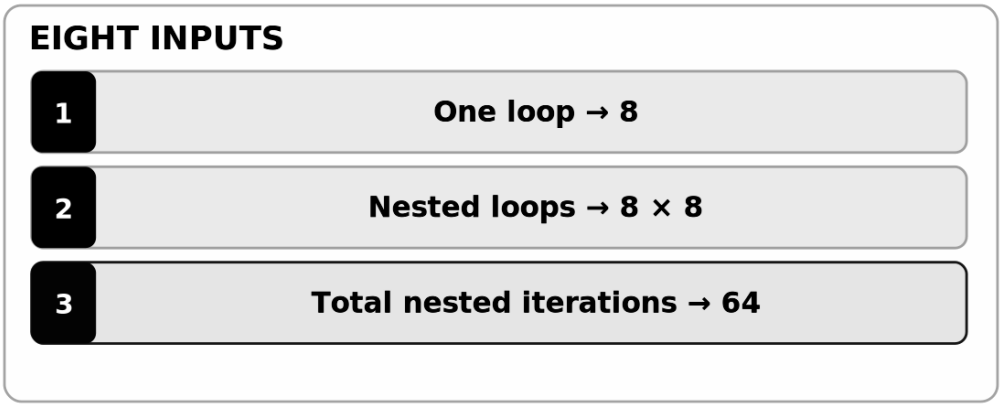
For n=8, compare one loop visiting each item with two full nested loops.

Follow the worked method

1. Predict first — One scan has eight iterations, giving linear growth O(n).

2. Explain the answer — Nested loops perform 8×8=64 inner iterations, giving quadratic growth $O(n^2)$.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first
The linear count doubles from 8 to 16.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

State the Big O of (a) constant assignment, (b) one n-element loop, (c) two full nested n-loops and (d) repeatedly halving an interval.

Question 2

A program copies an n-element array into a new n-element array. Give its time and auxiliary-space complexity and justify both.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

A method uses an in-place $O(n^2)$ sort and another uses merge sort with O(n log n) time and O(n) auxiliary space. Explain one situation favouring each.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

O(n) time necessarily means an algorithm uses O(n) additional memory.

Use key vocabulary from memory

1. Explain time complexity in your own words.

2. Explain auxiliary space in your own words.

3. Explain Big O in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.